

Do three thermometers tell the same story?

On-structure logger, town weather station and ERA5 reanalysis, 26 July 2018 to 13 August 2026

Study report · studies/proxy_comparison/

13 August 2026

Abstract

Three independent sources now measure the environment at Gubbio and the project has used only one of them. They share channel names and they do not measure the same quantity: a housing on a sun-exposed wall, a standard screen in town, and an average over roughly nine kilometres of Apennine terrain.

This comparison had to come before any of them is used to fill a gap. The study `unified_dataset` established that the on-structure channels vanish with the target — on 100 % of the hours the inclinometer is missing, so are its air temperature, humidity and battery — so an independent source is the only escape. Whether that escape is real depends on whether the proxy carries the same relationship with the wall.

It largely does, and the availability is decisive. During the hours the inclinometer is missing, the ground station is present on **94.3 %** and ERA5 on **99.6 %**. Against the on-structure sensor the ground station agrees to a mean absolute **1.12 °C** and ERA5 to **1.53 °C**, with correlations of 0.988 and 0.978.

One expectation was overturned. A gridded product was expected to under-represent the daily temperature swing. It does the opposite: ERA5’s diurnal amplitude is **19 % larger** than the on-structure sensor’s and the ground station’s is 15 % larger. The sensor closest to the wall is the most *damped* of the three, which is what a housing with thermal mass on a massive masonry wall should produce.

Agreement collapses under differencing. Correlation between sources falls from 0.99 in levels to **0.70** in first differences for temperature, and from 0.88 to 0.55 for humidity. The level agreement is largely a shared annual cycle; the hour-to-hour movement, which is what the wall responds to, agrees far less well.

A defect is confirmed by triangulation. The on-structure radiation channel agrees with neither proxy ($r \approx 0.74$), while the two proxies agree with each other at $r = 0.930$ and a slope of 0.998. Two independent instruments agreeing with each other and disagreeing with a third identify which one is wrong.

The relationship with the wall had to be measured inside contiguous blocks, and one defect had to be registered before it could be read at all. Computed on the whole record at once, as an earlier version of this report did, the differenced relationship appears negligible for every source and the on-structure radiation channel appears to *reverse* sign. Both are artefacts. Cut into blocks, kept inside one instrument era, and with 175 hours of single-hour acquisition faults registered and masked — hours in which the inclination, the air temperature and the battery all fail together and recover on the next sample — air temperature reaches $r = -0.95$ in levels and -0.84 in first differences against the calibrated inclination, and the radiation reversal disappears.

All three sources give the same sign, no channel leads the deformation, and every driver optimises at **zero transport delay**. Only solar radiation needs an operator: a one- to two-hour thermal inertia lifts its explained variance from 0.570 to **0.708**. Radiation adds one to three points of explained variance beyond air temperature — real, and small.

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1 What is being compared, and why

1.1 Three sources, one name

Suffix	Source	What it measures
_str	on-structure logger	Air inside the instrument housing on the monitored wall
_gs	ground station	Standard-exposure station in Gubbio town
_era5	ERA5 reanalysis	Average over a grid cell of roughly nine kilometres

They are not replicates. Disagreement between them is not error to be minimised; it is the physical difference between three measurement situations, and the study’s job is to measure it and say what it costs.

1.2 Why this had to come first

The deadlock this study exists to break

studies/unified_dataset/ established that every channel of the monitored station fails at once. On **100 %** of the hours the inclinometer is missing, the air temperature, humidity and battery are missing too. The missingness is outage-driven, so no method conditioned on an on-structure covariate can fill a real gap — the covariates are gone with the target.

An independent source does not share the station’s power supply. That is the only escape available, and whether it works depends on two things: is the proxy *there* when the target is not, and does it carry the *same relationship* with the wall. This study answers both.

1.3 Two rules

The target is fixed. The compensated inclination `inc_comp` is the calibrated reading: the manufacturer’s formula applied with the temperature measured at the transducer. That is an instrument calibration, not a regression. The transducer experiences the temperature at the transducer, so substituting a grid-cell or in-town temperature into the formula would be meaningless. Nothing in this study re-estimates the coefficient and no alternative target is built. That the on-structure temperature appears inside the target is a property of the calibrated measurement, stated here and not designed around.

The clock is measured, not assumed. The sources do not share a time convention, and an uncorrected offset would misplace every diurnal phase in the study.

2 Harmonisation, and the clock

2.1 What arrived

All three sources are placed on the hourly grid under one naming scheme, `{quantity}_{source}`, documented in `docs/proxy-data-dictionary.md` — which is generated from the mapping tables in `pc_lib` so that code and documentation cannot drift apart.

Two unit hazards were found and corrected. ERA5 reports relative humidity as a **fraction** where the other two report per cent; left uncorrected it would have appeared as a 99 % dry bias. The ground station is natively half-hourly with 124 duplicated timestamps.

Values outside a plausible physical range are **reported, not removed**. A radiation channel reading sixty watts at midnight is a finding about the sensor, and deleting it would hide the finding.

2.2 The radiation channels as measured

Every other figure in this report shows a statistic — a mean profile, an agreement scatter, a correlation. Figure 1 shows the measurements, and it is placed first because the strongest evidence against the on-structure pyranometer is visible in the raw channel before any statistic is computed.

The panels are restricted to the period the on-structure pyranometer covers, 21 February 2025 to 12 August 2026, because that is the only window over which the three sources can be compared at all. The proxies run from 2018; that difference in extent is what shapes the tiers in Section 4.3 and is reported there, and drawing it here would leave seven eighths of the on-structure panel empty and hide the comparison this figure exists to make.

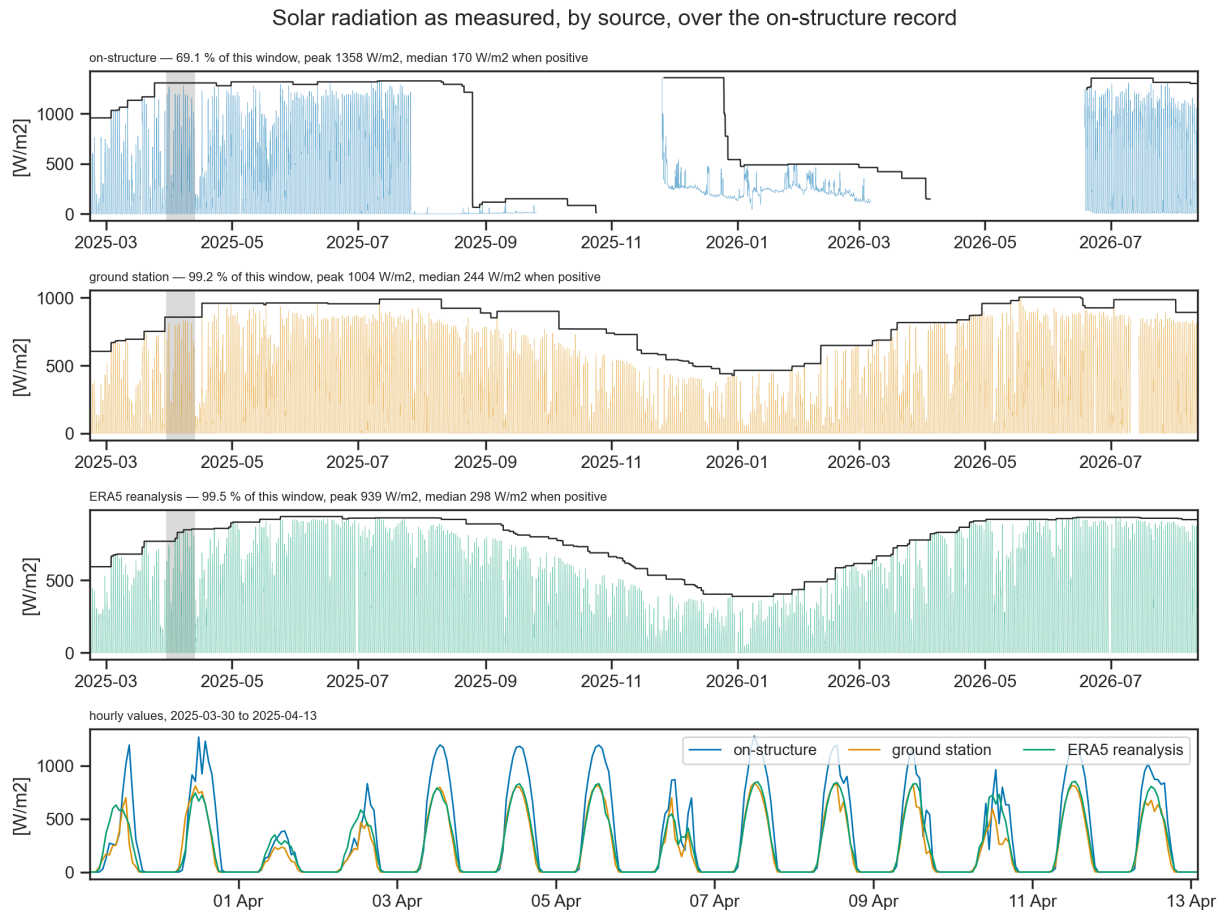


Figure 1: Solar radiation as measured, by source, over the on-structure record. The upper three panels give one source each at hourly resolution; the black line is a thirty-day rolling maximum, drawn as a seasonal ceiling rather than as a smoothing of the values beneath it. Hours with no measurement are left as breaks rather than joined, so an outage appears as the gap it is. The shaded band marks the fortnight enlarged below, chosen as the fourteen days in which all three sources are jointly most complete. The bottom panel overlays all three at full hourly resolution over that fortnight, which is the only scale at which an individual hourly value can be read.

The two proxies behave as a solar record should and the on-structure channel does not. The ground station and ERA5 panels each show one clean annual arc, a ceiling near 1000 W/m^2 in summer falling to roughly 450 in January and recovering, on 99.2% and 99.5% of the window respectively. The on-structure panel is present on **69.1%** of the same window and does none of that.

It over-reads whenever it works. Its peak is **1358 W/m^2** against 1004 for the ground station and 939 for ERA5, and in the enlarged fortnight its daily maxima run about **1.5 times** both proxies, day after day, in clear and overcast conditions alike. A clear-sky maximum above 1300 W/m^2 is not physically available at this latitude.

The channel fails outright for two months, and under-reads for four more

The monthly maxima make the failure explicit. Through the first summer the on-structure channel runs above both proxies, as it does again in the second: 1291, 1317 and 1326 W/m² in May, June and July 2025 against roughly 950 for each proxy, and 1353, 1312 and 1154 in June, July and August 2026.

In August and September 2025 it collapses. Its monthly maxima are **116** and **150** W/m² while both proxies report a normal late summer near **890**. A pyranometer that never exceeds 150 W/m² across an entire Umbrian August is not measuring sunlight.

From December 2025 to March 2026 it under-reads, inverting its usual bias: 541, 496, 462 and 354 W/m² against proxy maxima that rise from about 480 back to 816 as the season turns. By February and March it sits *below* both proxies, having been half again above them six months earlier.

This matters for the agreement statistics of Section 3. The 368-day overlap on which `str-gs` and `str-era5` are computed contains both a period of total failure and a period of inverted bias. Their correlations near $r = 0.74$ are therefore not the precision of a working instrument with an offset; they are an average over an instrument that was working, an instrument that was dead, and an instrument that was wrong in the opposite direction.

The two proxies track each other and the on-structure channel tracks neither. In the enlarged panel the ground station and ERA5 traces are nearly superimposed, including on the broken days of 30 March and 6 April, while the on-structure trace departs from both in amplitude on every day. That is the triangulation Section 3 quantifies, visible before any statistic is taken, and the conclusion it supports — that this channel is not usable as a physical measurement without recalibration — is reached here on the raw values alone.

2.3 The clock, settled two ways

Two independent tests are available and they do not have to agree.

The **absolute** test compares each source's solar-radiation phase against solar noon computed from the site's longitude and the equation of time. It needs no reference sensor. The **relative** test cross-correlates each source against the on-structure channel; it needs a reference but works on any quantity.

Source	Season	offset from solar noon	shift on tair	shift on sr
on-structure	winter	+0.77 h	—	—
on-structure	summer	+1.26 h	—	—
ground station	winter	+0.09 h	0 h	1 h
ground station	summer	+0.06 h	1 h	1 h
ERA5	winter	+0.51 h	0 h	0 h
ERA5	summer	+0.42 h	1 h	1 h

The on-structure radiation channel fails its own consistency check

It places solar noon *after* its own daily temperature maximum, which is impossible — the wall cannot warm before the sun reaches its highest point. Its winter estimate has an interquartile range of **5.3 hours**, and it reads roughly **60 W/m² at midnight**.

The alignment is therefore taken from **air temperature**, which all three sources carry with good coverage and which is the quantity the rest of the study uses. On temperature the answer is unambiguous: the ground station and ERA5 agree with each other *exactly* in both seasons, and both sit one hour from the on-structure logger **in summer only**.

That is the signature of a daylight-saving mismatch — one clock advances in spring and the other does not — so the correction is applied by season. A single annual figure would have been wrong in both halves of the year. After correction the residual offset is zero on every source and season, which the notebook asserts.

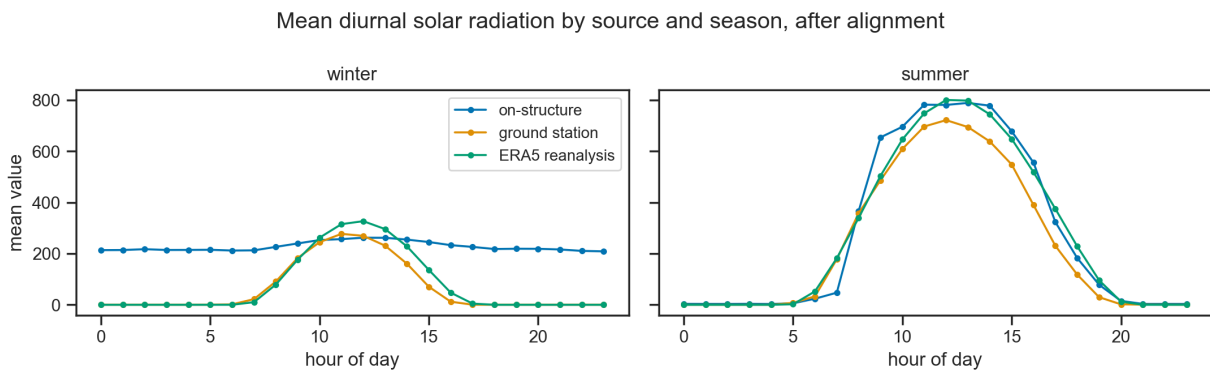


Figure 2: Mean diurnal solar radiation by source and season after alignment. The on-structure channel’s non-zero night-time floor is visible directly.

3 Absolute agreement

Three complementary views, because they answer different questions. Bias and RMSE say how large the disagreement is. The regression slope and intercept say what *kind* it is — a slope near one with a non-zero intercept is an offset, a slope away from one is a scale error, and they call for different remedies. The limits of agreement say how far a single substituted value might fall from the one it replaces.

pair	days	bias	MAE	slope	r	limits of agr.
<i>air temperature [$^{\circ}\text{C}$]</i>						
str-gs	1958	+0.37	1.12	0.942	0.988	−2.3 to +3.0
str-era5	2048	−0.57	1.53	0.922	0.978	−4.2 to +3.0
gs-era5	2772	−0.92	1.36	0.969	0.981	−4.0 to +2.1
<i>relative humidity [%]</i>						
str-gs	1926	−6.51	8.57	0.775	0.880	−26 to +13
str-era5	2048	−1.28	9.41	0.686	0.822	−25 to +22
gs-era5	2690	+5.80	8.80	0.745	0.806	−17 to +28
<i>solar radiation [W/m^2]</i>						
str-gs	368	−67.0	153.9	0.577	0.744	−505 to +370
str-era5	368	−43.1	153.1	0.606	0.738	−489 to +403
gs-era5	2774	+29.1	47.4	0.998	0.930	−152 to +211
<i>solar radiation, daylight hours only</i>						
str-gs	194	−48.8	211.5	0.489	0.705	−600 to +503
str-era5	195	−7.3	210.9	0.494	0.700	−564 to +549
gs-era5	1395	+54.1	90.0	0.890	0.886	−190 to +298

Temperature substitutes well; humidity does not

Air temperature is the one channel a proxy can stand in for. The ground station tracks the on-structure sensor to a mean absolute **1.12 $^{\circ}\text{C}$** at $r = 0.988$, and ERA5 to 1.53 $^{\circ}\text{C}$ at $r = 0.978$. The ground station is the closer of the two on every measure, as its position should imply.

Humidity is not interchangeable. Correlations of 0.81–0.88 and limits of agreement spanning **fifty percentage points** mean a substituted humidity value carries no useful precision. Any use of humidity as a regressor must come from the source that will actually be available, not be swapped between them.

The radiation defect, confirmed by triangulation

The on-structure radiation channel agrees with neither proxy: $r = 0.744$ against the ground station, 0.738 against ERA5, with mean absolute errors above 150 W/m^2 and limits of agreement spanning nearly nine hundred.

The two proxies, meanwhile, agree with *each other* at $r = 0.930$, a slope of **0.998** and a mean absolute error of 47 W/m^2 — a third of the disagreement either has with the on-structure sensor, over seven times the overlap.

Two independent instruments agreeing with each other and disagreeing with a third identify which one is wrong. Combined with the night-time floor and the impossible solar-noon phase of Section 2, the on-structure radiation channel should not be used as a physical measurement without recalibration.

Half of a radiation record is night, and it flatters every statistic

Roughly half of every solar-radiation series is a structural night-time zero. Those zeros are real measurements and are removed nowhere in this study, but a statistic computed over the whole day is computed largely on a block of samples on which all three sources agree trivially, because all three are reporting the same zero.

Restricting to the hours the sun stands above the horizon — a geometric mask taken from the site coordinates, independent of all three sources — removes that flattery, and the effect is large. The correlation between the two proxies falls from **0.930 to 0.886** and their mean absolute difference nearly doubles, from **47.4 to 90.0 W/m²**; their regression slope falls from 0.998 to 0.890. The two comparisons involving the on-structure channel move less, from $r \approx 0.74$ to $r \approx 0.70$, because that channel disagrees with both proxies during the day as well as at night.

The daylight figures are the ones that describe how well a proxy could stand in for a radiation measurement, since a reconstruction is only ever tested where there is something to reconstruct. The all-hours figures are kept beside them because they are what a naive comparison reports.

The mask is geometry, not radiation. It takes no account of cloud, and none of the local horizon formed by the terrain around the site, so it marks when the sun is up rather than when the wall was lit.

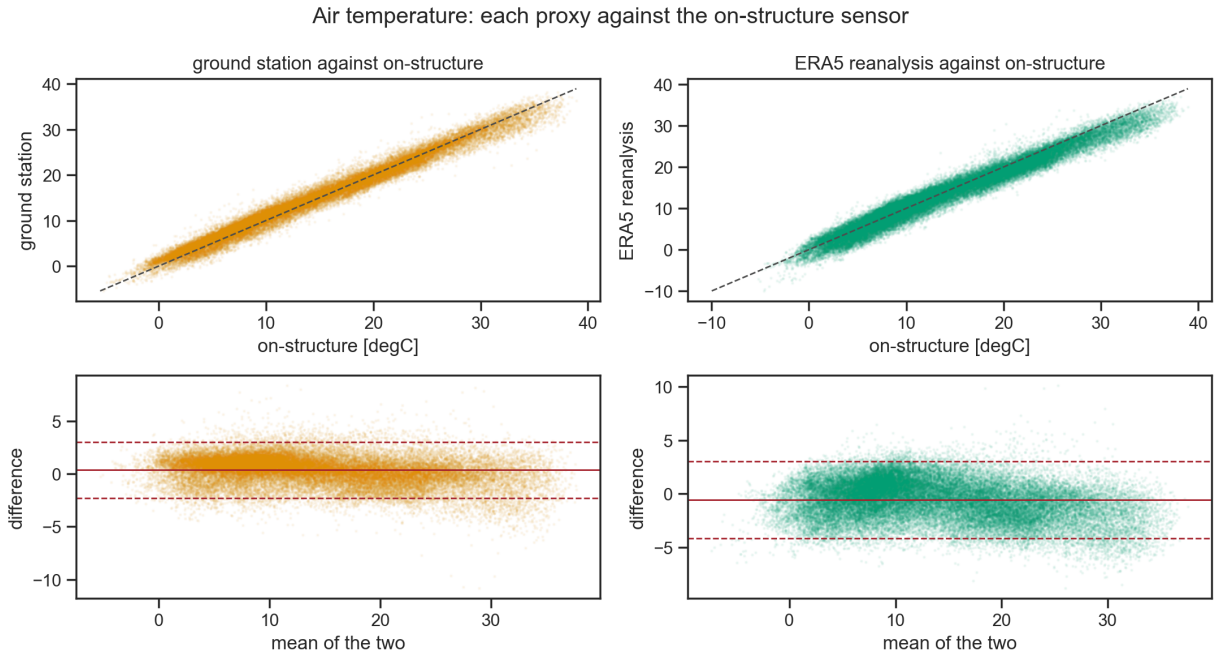


Figure 3: Air temperature. Top: each proxy against the on-structure sensor, with the identity line. Bottom: Bland–Altman, difference against mean, with the mean difference and the limits of agreement.

4 Dynamic agreement

A proxy can sit two degrees low and still drive the wall correctly, or match on average and miss every daily swing.

4.1 The daily cycle

Source	diurnal amplitude	ratio to on- structure	peak hour
<i>air temperature [°C]</i>			
on-structure	3.31	1.00	13
ground station	3.80	1.15	13
ERA5	3.94	1.19	14
<i>relative humidity [%]</i>			
on-structure	10.22	1.00	7
ground station	11.87	1.16	6
ERA5	13.91	1.36	5
<i>solar radiation [W/m²], daylight hours</i>			
on-structure	291.9	1.00	12
ground station	243.0	0.83	12
ERA5	282.6	0.97	12

The radiation rows are computed over the daylight hours and carry no trough hour. Over the full day the trough of a radiation cycle is an arbitrary point inside the night — the all-hours computation returns hours 23, 1 and 2 for the three sources, which are three different ways of saying “dark” — and a number that arbitrary does not belong in a comparison. All three sources peak at hour 12, which is the one phase statement the radiation channels agree on.

Radiation is also the one quantity where the on-structure sensor shows the *largest* daily amplitude rather than the smallest. That is consistent with the defect established above rather than with the damping seen in temperature: a channel reading roughly 60 W/m² at midnight and disagreeing with both proxies has no claim to a physically meaningful amplitude.

The expectation was that ERA5 would flatten the daily cycle. It does the opposite

A nine-kilometre grid average should smooth a microclimate away, so the reanalysis was expected to show the *smallest* diurnal amplitude. It shows the **largest** — 19% above the on-structure sensor, with the ground station 15% above.

The sensor closest to the wall is the most **damped** of the three, and on reflection that is what the physics requires. The on-structure channel measures air inside an instrument housing mounted on a massive masonry wall. The housing has thermal mass and the wall behind it has a great deal more, so the air the sensor sees is buffered against the daily swing that a freely ventilated screen in a field records in full.

The consequence for imputation is favourable and was not anticipated. A proxy that under-represented the daily cycle would have been unable to reconstruct it. Both proxies over-represent it instead, which is a scale difference — correctable by regression — rather than missing information.

ERA5’s temperature peak falls one hour later than the other two. The humidity peaks spread over three hours, which is a further reason not to substitute humidity between sources.

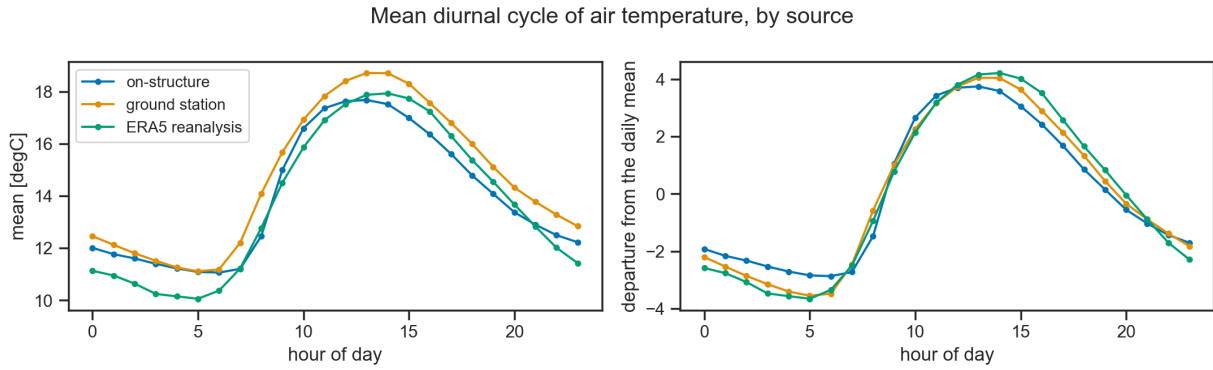


Figure 4: Mean diurnal cycle of air temperature by source. Left: absolute. Right: departure from each source’s own daily mean, which isolates the shape.

4.2 What survives differencing

The wall responds to change, not to level, and the first difference is the quantity invariant to every arbitrary constant in the chain.

pair	r levels	r differences	lost	slope
<i>air temperature</i>				
str-gs	0.988	0.699	-0.29	0.684
str-era5	0.978	0.690	-0.29	0.639
gs-era5	0.981	0.731	-0.25	0.679
<i>relative humidity</i>				
str-gs	0.880	0.546	-0.33	0.542
str-era5	0.822	0.489	-0.33	0.498
<i>solar radiation</i>				
str-gs	0.744	0.491	-0.25	0.343
gs-era5	0.930	0.648	-0.28	0.537

Most of the level agreement is a shared annual cycle

Differencing costs about **0.29** of correlation on every temperature pair and about 0.33 on humidity. Two sources correlating at 0.99 in levels agree only at 0.70 on what happened between one hour and the next.

This is not a defect in either instrument. Two thermometers a few kilometres apart share the season and the synoptic weather almost perfectly, and differ on the local gust, the passing cloud and the sensor’s own response time — all of which live in the difference.

It sets the ceiling for proxy-driven reconstruction. A model driven by a proxy can recover the part of the wall’s movement that the proxy shares with the on-structure sensor, and that share is far smaller in the domain the wall actually responds to. The slopes below one — 0.64 to 0.68 for temperature — are partly regression dilution, since the reference carries measurement noise of its own, so they bound rather than measure the attenuation.

Every statistic up to this point was computed on the joined frame: 26 July 2018 to 13 August 2026, both instrument eras, and ragged throughout. No correlation against the target is computed that way below, for two reasons that this section establishes before it reports anything.

4.3 The windows this comparison can use

The prediction studies never correlate against the target on a span of that kind. They cut contiguous blocks of near-complete coverage, group them into tiers according to which channels a tier requires, and report every correlation, slope and operator inside a tier. This report now does the same, with one addition the sibling studies do not need: because it spans both eras, it must also keep them apart. The instrument at `st02` was physically reinstalled on 21 February 2025 and the two packages share no baseline, so no window here crosses that date.

Tier	Requires	Blocks	Longest block	Total days
P0	target and all three sources	7	86.6 d	343.4
P1	target and both proxies	7	86.6 d	386.4
P2	target and both proxies, legacy era	23	107.8 d	1271.2

Blocks are contiguous runs in which every required channel is present, interruptions up to six hours absorbed, minimum length twenty days — the same parameters the prediction studies use, so the inventories are comparable. Full listing in `P_13`.

The radiation record sets the shape of the tiers by itself. `sr_str` begins on 21 February 2025, which is the era boundary, so a tier requiring the on-structure pyranometer cannot reach the legacy era at all. The three-source radiation comparison therefore lives entirely inside the current era, while the two proxies additionally carry a legacy era the on-structure channel can never enter. That is the whole of what “solar radiation is available for only a fraction of the period” means here, and it is a property of the tiers rather than a caveat attached to them.

Tiers P0 and P1 differ only in whether the on-structure channels are required. P1 contains every hour of P0 and a little more, so the two are not independent samples and their agreement is weaker evidence than it looks. Where they differ at all below, the difference is one block.

4.4 A defect that had to be settled first

Blocking alone did not reconcile this report with the prediction studies, and the reason turned out to be in the record rather than in the method.

Single-hour acquisition faults, and why they had to be registered

The current era contains hours in which the calibrated inclination moves by hundreds of millidegrees and returns on the following sample. At **2025-05-10 14:00** the raw inclination moves -220 mdeg, the air temperature rises 5.9°C and the battery voltage dips — all within the same hour, all recovering at the next. Three physically independent channels do not fail together for one sample because a wall moved.

The compensation is not the cause and differencing is not the cause. The raw channel already carries 110 such hours in the current era against 47 in the legacy era; the calibrated channel carries 166 against 37, inheriting them and enlarging those where the temperature spiked as well. The index is a complete hourly grid, so no difference is ever taken across a gap.

They matter because they live almost entirely in the first difference: one unreversed hour contributes two large differences of opposite sign. Left in place they do not merely add noise — they reverse conclusions.

The hours are marked, not deleted. A rolling median and a median absolute deviation over a one-day half-window score every hour, and those beyond eight robust standard deviations are entered in a register, `P_23`, together with the evidence for each: the signed step, how many other channels were simultaneously anomalous, and whether the excursion reversed on the next hour. The register holds **175 hours, 0.248 %** of the grid. The mask is applied to the *level*, so one decision governs the level statistics and the differenced ones rather than two separate rules, and a sustained shift — a real re-levelling — is not caught by it.

	<i>r</i> unscreened	<i>r</i> screened	sibling study
<i>tier P0, on-structure air temperature</i>			
levels	−0.757	−0.952	−0.956
differences	−0.148	−0.840	−0.886
<i>tier P0, on-structure solar radiation</i>			
differences	+0.036	−0.533	−0.624
<i>tier P2, on-structure air temperature</i>			
differences	−0.890	−0.890	−0.901

Sibling figures from `studies/inclination_prediction` for the current era and `studies/inclination_prediction_legacy` for the legacy era.

The register changes the current era and leaves the legacy era alone. Every legacy figure moves by 0.002 or less, which is why tier P2 reproduced its sibling study before any screening at all; the legacy faults exist but are fifty times smaller, a maximum step of 51 mdeg against 1314. In the current era the correction is decisive in both domains, and it recovers the sibling studies' values from a record that had appeared to contradict them.

A reversal that was not a defect

Unscreened, the on-structure radiation channel correlates **positively** with the target in first differences, $r = +0.036$, while both proxies stay negative. Read on its own that is a fourth symptom of a channel already known to be defective.

It is not. Screened, the same channel gives $r = -0.533$, in agreement with both proxies and with the sibling study. **The apparent sign reversal was produced entirely by the acquisition faults**, and nothing in this report reopens the sign question of docs/raw-data-format.md §7.5.

4.5 Correlation structure

With the windows defined and the faults registered, the correlation matrix over the target and every channel a tier carries, on levels and on first differences.

	on- structure	ground station	ERA5
<i>air temperature</i>			
P0 levels	−0.952	−0.940	−0.932
P0 differences	−0.840	−0.582	−0.570
P2 levels	−0.541	−0.543	−0.539
P2 differences	−0.890	−0.668	−0.569
<i>solar radiation</i>			
P0 levels	−0.142	−0.442	−0.460
P0 differences	−0.533	−0.430	−0.527
P2 levels	—	−0.222	−0.215
P2 differences	—	−0.491	−0.580

Correlations between two trending series are inflated by the shared trend and say little about a driver–response relationship; the differenced matrix is the honest one for a signal with drift. Both are shown so the gap is visible, which is the convention the prediction studies use.

The two eras answer differently, and the difference is instructive. In the current era the level correlation is the stronger one for temperature and the differenced correlation the weaker; in the legacy era it is the reverse. The legacy blocks are drawn from seven calendar years and no single block spans an annual cycle, so their levels carry less common trend to inflate them, while their hour-to-hour movement is measured on a quieter instrument.

Radiation is the one quantity that gains under differencing in both eras. Its level correlation with the target is weak everywhere — between -0.14 and -0.46 — and its differenced correlation is stronger for every source in every tier. Radiation drives the daily cycle and carries almost none of the seasonal level, which is exactly the signature differencing exposes.

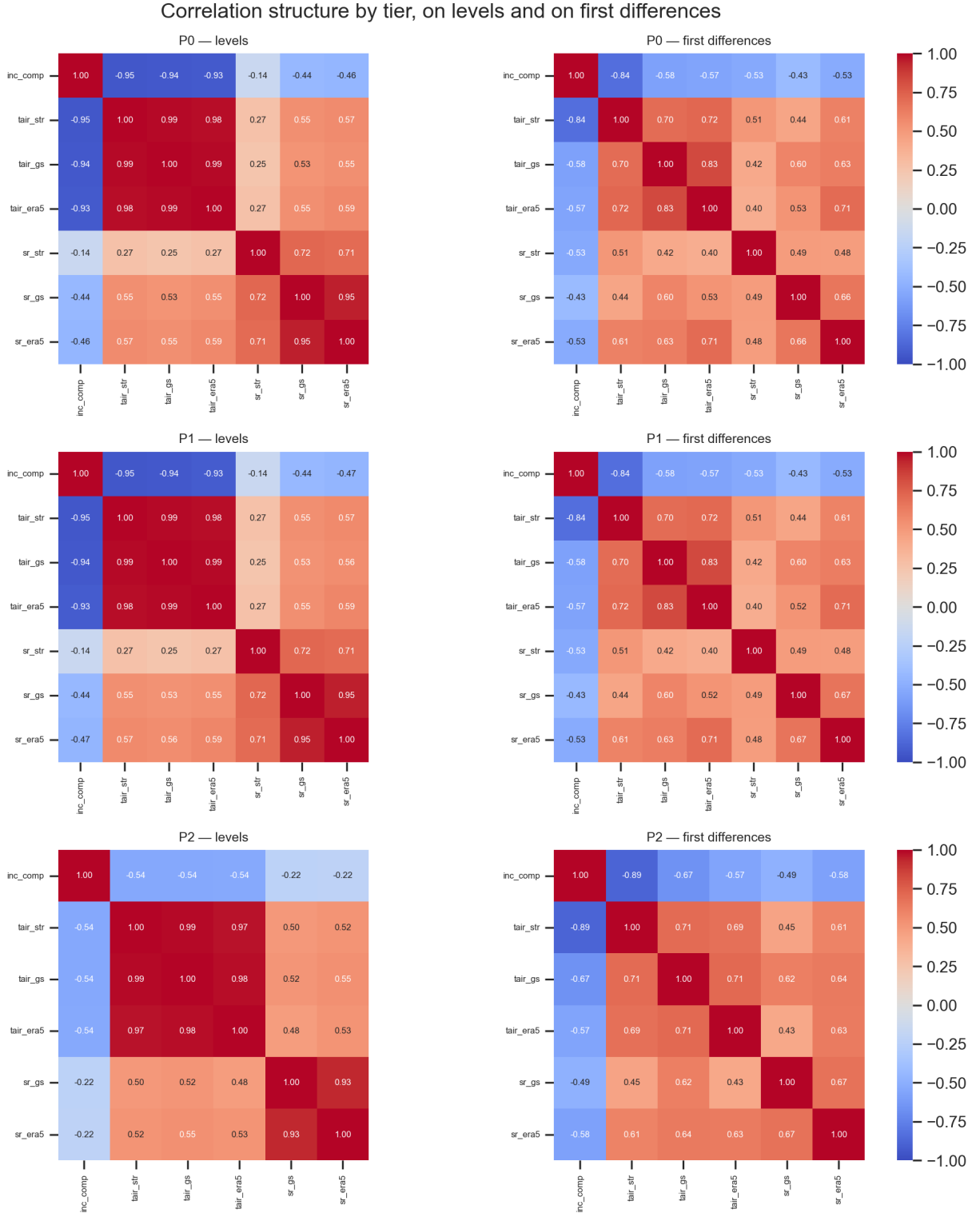


Figure 5: Correlation structure by tier, on levels (left) and on first differences (right). The block of high mutual correlation among the drivers themselves is quantified in Section 4.9.

4.6 Relationship with the target, by tier

Each source against the one target, in both domains, inside each tier’s blocks. The slope describes the residual response that survives the calibration; **it is not a candidate compensation coefficient**.

Source	r	slope [mdeg unit ⁻¹]	s.e.	R^2
<i>air temperature [mdeg °C⁻¹], tier P0, levels</i>				
on-structure	−0.952	−4.602	0.067	0.907
ground station	−0.940	−4.842	0.074	0.883
ERA5	−0.932	−4.766	0.074	0.868
<i>air temperature, tier P0, first differences</i>				
on-structure	−0.840	−3.010	0.041	0.706
ground station	−0.582	−2.391	0.050	0.339
ERA5	−0.570	−2.336	0.051	0.325
<i>solar radiation [mdeg (W m⁻²)⁻¹], tier P0, first differences</i>				
on-structure	−0.533	−0.0167	0.0006	0.284
ground station	−0.430	−0.0187	0.0008	0.185
ERA5	−0.527	−0.0276	0.0008	0.278
<i>air temperature, tier P2, first differences</i>				
on-structure	−0.890	−2.653	0.020	0.792
ground station	−0.668	−1.966	0.024	0.447
ERA5	−0.569	−1.766	0.023	0.323

Standard errors are Newey–West at 24 lags. The full table, every quantity, every tier, both domains, and both the screened and the unscreened frame, is P_16.

Every source agrees on the sign, and the proxies pay a measurable price

All twelve correlations above are **negative**, which is the direction the site’s geometry requires: the valley face is the sun-exposed one, it expands more under heating, and the wall tips toward the mountain (docs/raw-data-format.md §7.5). No source and no tier reverses it.

The on-structure sensor retains substantially more of the differenced relationship than either proxy — −0.840 against −0.582 and −0.570 in the current era, −0.890 against −0.668 and −0.569 in the legacy era. That gap is the price of substitution, and it is the quantity a reconstruction study inherits: a proxy-driven model can recover the part of the wall’s movement that the proxy shares with the wall’s own sensor, and in the difference domain that share is roughly two thirds.

Solar radiation reaches roughly half the differenced correlation of air temperature, −0.53 against −0.84 in tier P0, from sources that agree with each other far better than the radiation channels do. It is a real driver in the domain the wall responds to, not a residual.

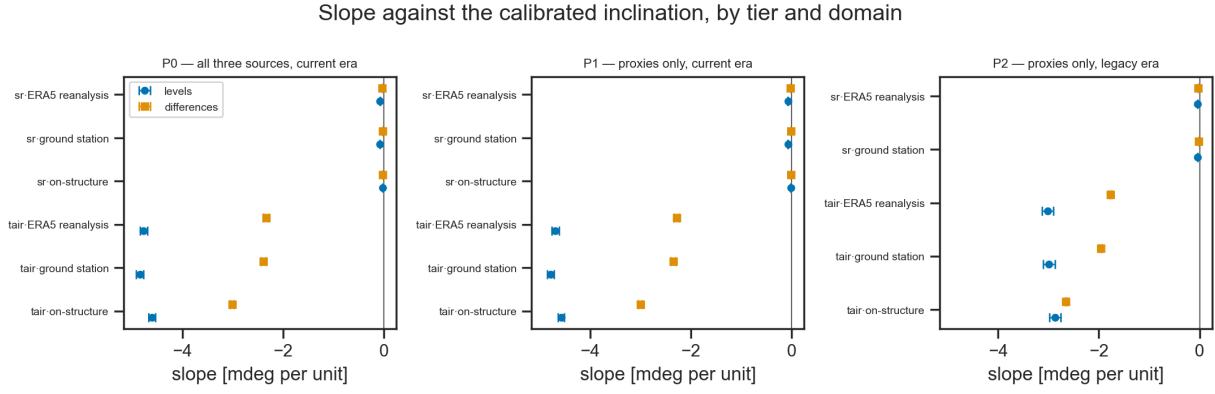


Figure 6: Slope against the calibrated inclination, by tier and domain, for air temperature and solar radiation. Error bars are Newey–West standard errors at 24 lags.

4.7 Transport delay and thermal inertia, on two bands

Before a driver is compared it is given the chance to act through the operator that suits it. A **transport delay** shifts the driver without changing its shape; a **thermal inertia** low-passes it through a single-pole filter. The two are scanned jointly over 13 delays and 16 time constants, because scanning either alone can attribute to one what belongs to the other, and the instantaneous case stays in the grid so that it competes rather than being assumed away.

The scan runs on each tier’s *longest single block*, not on the union of its blocks. A delay is applied with a shift, a time constant with a recursive filter and the band limit with a rolling mean; all three read across adjacent rows, so a union would let a filter draw values from the far side of a months-long interruption.

Tier	Driver	delay	τ	r	R^2	R^2 at $d = \tau = 0$
<i>diurnal band — carried forward</i>						
P0	tair_str	0 h	0 h	−0.977	0.955	0.955
P0	tair_gs	0 h	0 h	−0.930	0.865	0.865
P0	tair_era5	0 h	0 h	−0.916	0.839	0.839
P0	sr_str	0 h	1 h	−0.542	0.294	0.262
P0	sr_gs	0 h	2 h	−0.841	0.708	0.570
P0	sr_era5	0 h	1 h	−0.829	0.686	0.618
P2	tair_str	0 h	0 h	−0.964	0.930	0.930
P2	sr_gs	0 h	2 h	−0.746	0.556	0.437
P2	sr_era5	0 h	1 h	−0.711	0.506	0.460

Delays bounded at twelve hours per `docs/raw-data-format.md §7.5`: both air temperature and solar radiation are external forcings, and beyond half a diurnal cycle a delay is indistinguishable from a lead. Tier P1 reproduces P0 exactly, because the two share their longest block. Full-band results and the complete table are P_17 and P_18.

No driver needs a transport delay; only radiation needs an inertia

Every optimum sits at zero delay, on both bands and in both eras. A grid cell averaged over nine kilometres was expected to show a different apparent lag from a sensor on the wall; it does not. A proxy therefore needs no phase correction before use, which is a simplification the scan earned rather than an assumption.

Air temperature also needs no inertia — its optimum is $\tau = 0$ for every source in every tier, and the operator adds nothing to the instantaneous fit.

Solar radiation does. Both proxies optimise at a one- or two-hour time constant, and the gain is not marginal: for the ground station explained variance rises from **0.570 to 0.708**. Radiation heats a surface which then warms the air and the masonry, so a short lag between the forcing and the response it produces is what the physics predicts. This is the one place in the report where fitting an operator changes a conclusion.

The full band produced no boundary optimum. The prediction study found solar radiation pinned at the last time constant in the grid, 168 hours, which it correctly diagnosed as a filter turning a daily cycle into a seasonal envelope. On these blocks that does not occur for any driver: no full-band optimum sits at the grid edge. The blocks here are shorter than an annual cycle, so there is no seasonal component for a week-wide filter to reach for.

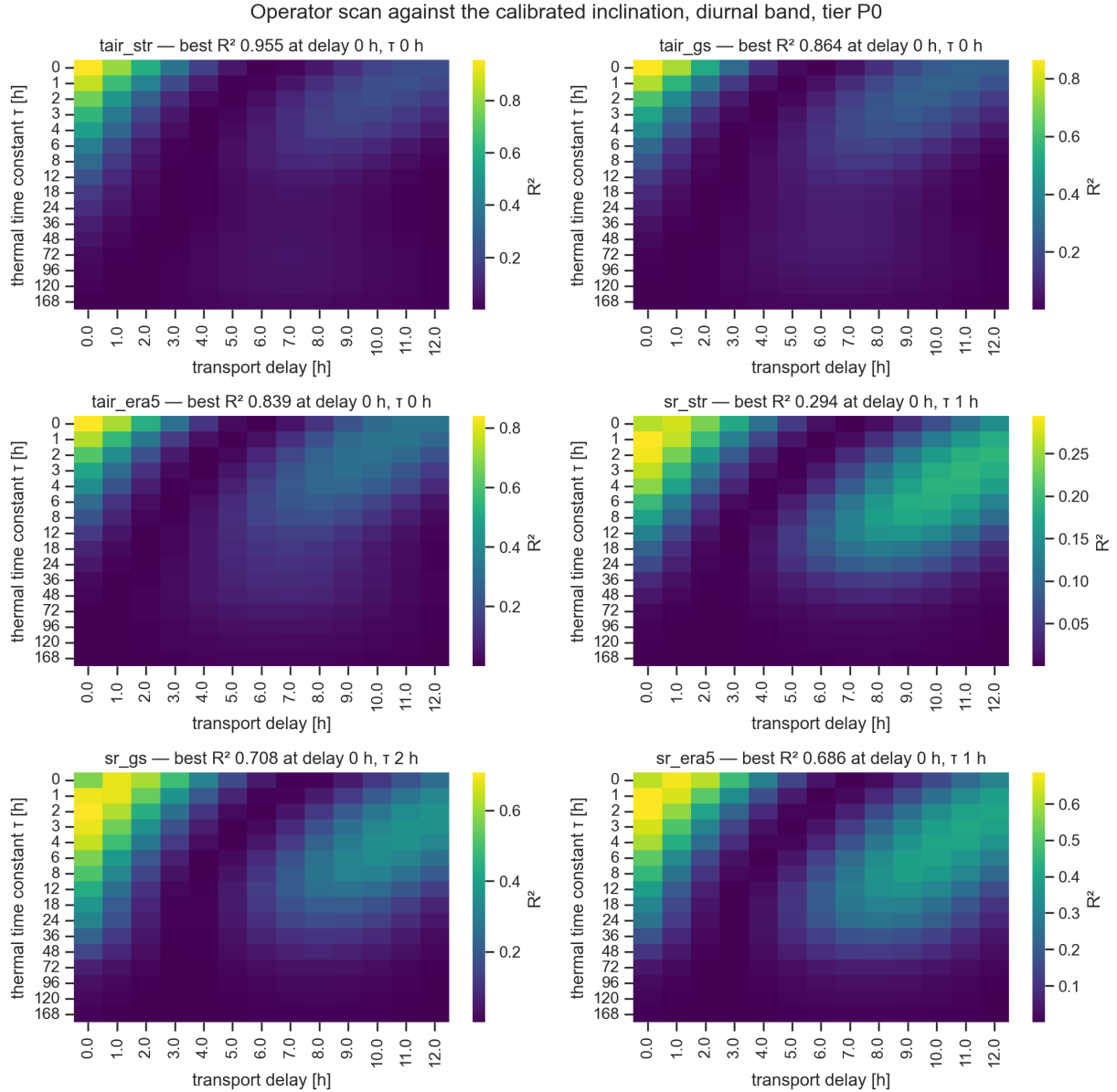


Figure 7: Explained variance over the delay-by-inertia grid, diurnal band, tier P0. Every optimum sits on the zero-delay edge; the radiation panels brighten one to two hours into the inertia axis, which is the operator the table reports.

4.8 Does any radiation channel lead the response?

The scan above is bounded at non-negative delays because a forcing must precede the response it causes. That bound is right for *use* — a lead cannot be applied by a forecasting system, which would need the driver’s future values at the moment the forecast is issued — but it also hides a diagnostic. If an optimum genuinely sat at a negative delay, the bounded scan would report the boundary instead of the anomaly. Scanning signed delays makes it visible.

Tier	Channel	peak lag	r at peak	r at lag 0
P0	<code>sr_str</code>	+1 h	−0.524	−0.512
P0	<code>sr_gs</code>	+1 h	−0.825	−0.755
P0	<code>sr_era5</code>	+1 h	−0.821	−0.786
P2	<code>sr_gs</code>	+1 h	−0.721	−0.661
P2	<code>sr_era5</code>	+1 h	−0.700	−0.679

Every radiation channel peaks at a positive lag of one hour, in both eras and including the defective on-structure channel. A positive lag means the driver leads the response, which is the only direction a forcing can act in, and one hour is consistent with the one- to two-hour inertia the operator scan selected independently. No channel leads the deformation, so the causal constraint imposed on every model in this project costs nothing here, and the sign anomaly of Section 4.4 is confirmed as an artefact of the acquisition faults rather than a property of any instrument.

4.9 What radiation adds once temperature is present

Radiation heats the air, so a marginal correlation between radiation and the target partly re-measures the temperature relationship. The variance inflation factor quantifies the duplication, and a joint fit reports what radiation adds beyond temperature alone.

Tier	Source	VIF	r between	R^2 tair	R^2 joint	ΔR^2
<i>first differences</i>						
P0	on-structure	1.37	0.520	0.706	0.719	+0.013
P0	ground station	1.57	0.602	0.339	0.349	+0.010
P0	ERA5	2.01	0.709	0.325	0.355	+0.031
P2	ground station	1.64	0.626	0.447	0.456	+0.009
P2	ERA5	1.67	0.634	0.323	0.404	+ 0.081

Newey–West standard errors at 24 lags. The radiation term clears $|t| = 6$ in every row above and reaches $|t| = 45$ in the last. Level-domain rows are in P_20.

Radiation earns a place, and it is a small one

Collinearity is mild. Variance inflation factors run from 1.37 to 2.01, so the two drivers are not interchangeable and a joint fit is interpretable.

The increment is statistically overwhelming and practically small. Every radiation term clears $|t| = 6$, which on tens of thousands of hourly samples establishes only that the effect is not zero. What it buys is one to three points of explained variance in the current era, and eight in the legacy era for ERA5.

The recommendation follows the size, not the significance. A radiation channel is worth carrying as a second regressor where it is free — both proxies supply one — but it does not change what a temperature-driven model can do, and it is not a substitute for temperature anywhere.

4.10 What the pooled span cost

The previous version of this section reported the relationship on the whole record at once. Against the tiered, screened figures:

	pooled span	tier P0
<i>air temperature, first differences</i>		
on-structure	−0.155	− 0.840
ground station	−0.091	−0.582
ERA5	−0.084	−0.570
<i>solar radiation, first differences</i>		
on-structure	+0.087	− 0.533
ground station	−0.088	−0.430
ERA5	−0.091	−0.527

The earlier conclusion was an artefact of the analysis window and the unregistered faults, not a finding about the wall. That version reported the differenced relationship as weak for every source, with R^2 between 0.007 and 0.024, and read it as agreement with the prediction studies that the drivers contribute nothing measurable. The prediction studies report no such thing: on their own blocks they find air temperature at $r \approx -0.89$ in first differences, which is what this report now also finds. The pooled figures are retained in P_08 as the contrast, and are not interpreted anywhere.

5 Verdict

5.1 Availability, which decides everything else

Source	coverage of archive span	present when the target is missing
<i>air temperature</i>		
on-structure	70.1 %	0.0 %
ground station	94.8 %	94.3 %
ERA5	99.4 %	99.6 %
<i>solar radiation</i>		
on-structure	12.7 %	0.0 %
ground station	94.9 %	94.4 %
ERA5	99.4 %	99.6 %

The deadlock is broken

The on-structure sensor is present on **none** of the hours the target is missing — that is what outage-driven missingness means, and it is why the previous study had to refuse the driver-based filler.

The ground station is present on **94.3 %** of those hours and ERA5 on **99.6 %**. The availability criterion that emptied the candidate pool is satisfied by both proxies, and comfortably.

Radiation behaves the same way, and the asymmetry is extreme. The on-structure pyranometer covers **12.7 %** of the archive span and, like every on-structure channel, none of the hours the target is missing. Both proxies supply radiation on more than 94 % of exactly those hours. A radiation regressor is therefore available for reconstruction only if it comes from a proxy, which is the same conclusion temperature reached and for the same reason.

5.2 Which source, for what

Use	Recommendation
Gap reconstruction	ERA5 as primary, at 99.6 % availability, with the ground station as a cross-check where both exist. ERA5’s slightly worse agreement (1.53 against 1.12 °C) is outweighed by its near-total coverage.
Regressor for a forecast model	Either , with ERA5 preferred for continuity. Both need a scale correction: their diurnal amplitude runs 15–19 % above the wall’s own sensor.
Compensation	On-structure only , always. The calibration is an instrument correction and must use the temperature at the transducer.
Humidity, any use	Do not substitute . Limits of agreement span fifty percentage points.
Radiation as a second regressor	Either proxy , applied through a one- to two-hour inertia filter. It adds one to three points of explained variance over air temperature alone, which is worth taking where it is free and is not worth designing around.
Radiation, as a measurement	Not the on-structure channel until it is recalibrated. The two proxies agree with each other and both disagree with it, and it reaches barely half their differenced correlation with the target.

5.3 The feature specification for the imputation study

Channel	Source	delay	τ	R^2 diurnal	r diff.	available in gaps
tair_str	on-structure	0 h	0 h	0.955	−0.840	0.0 % fail
tair_gs	ground station	0 h	0 h	0.865	−0.582	94.3 % pass
tair_era5	ERA5	0 h	0 h	0.839	−0.570	99.6 % pass
sr_str	on-structure	0 h	1 h	0.294	−0.533	0.0 % fail
sr_gs	ground station	0 h	2 h	0.708	−0.430	94.4 % pass
sr_era5	ERA5	0 h	1 h	0.686	−0.527	99.6 % pass

Operators from the diurnal-band scan on tier P0’s longest block; differenced correlations from tier P0. Full table in P_12.

Both proxies enter the imputation study **unlagged**, which is a simplification the operator scan earned rather than an assumption. Their temperature channels enter without an inertia filter as well; their **radiation channels do not** — a one- to two-hour time constant raises explained variance for the ground station from 0.570 to 0.708 and is the one operator in this report that changes a conclusion.

6 Caveats

Agreement is not interchangeability. Two temperature series can agree to 1.1 °C and still imply different things for a wall, because the wall responds to the difference and the difference agrees far less well.

The differenced slopes are attenuated. Regression dilution biases them toward zero whenever the reference carries noise of its own, which it does. They bound the attenuation rather than measuring it.

The radiation comparison rests on a short and partly broken overlap. On-structure radiation exists only from 2025-02-21, giving 368 days against the 2774 available for the two proxies, and 194 days once the daylight mask is applied. Section 2.2 shows that the channel failed outright for two months of that overlap and under-read for four more, so every statistic involving it pools three different instrument states and should be read as a description of the record rather than of a sensor.

The legacy tier measures a different instrument. Tier P2 lies entirely before the 21 February 2025 reinstallation. The package installed on that date shares no baseline with the one it replaced, so a result obtained on P2 describes the earlier instrument. No window in Section 4.2 spans the boundary, and no figure is carried across it.

Tiers P0 and P1 overlap. P1 contains every hour of P0 and one further block. Agreement between the two is therefore weaker evidence than agreement between independent samples, and the two differ nowhere that matters in this report because they share their longest block.

The daylight mask is geometry, not radiation. It marks the hours the sun stands above the horizon at the site’s coordinates. It takes no account of cloud, and none of the local horizon formed by the terrain around the site, so it says when the sun was up rather than when the wall was lit.

The fault register is a threshold, and thresholds can be argued with. Eight robust standard deviations is deliberately conservative; the count of registered hours changes with it, though the conclusions do not over the range tested. The register is published as P_23 with the evidence for every flagged hour precisely so that the judgement can be inspected rather than taken on trust. The faults themselves are a property of `data/interim/unified/`, which every study in this project reads, and correcting them at that source rather than in this study would be the more durable fix.

One station. All three sources are compared at `st02` only, by instruction and for comparability with the prediction studies.

The clock correction is empirical. It is measured on temperature and applied by season. It reconciles the sources to within one hour, which is the grid resolution; a sub-hourly residual would not be visible.

7 Artefact inventory

Artefact	Content
P_01 – P_03	Channel inventory, clock test, comparable channels with overlaps
P_04, P_05	Pairwise agreement; the same split by season and hour
P_06, P_07	Diurnal amplitude and phase; first-difference agreement
P_08, P_10	Relationship with the target on the pooled span, retained as the contrast case; pooled slopes
P_11, P_12	Verdicts; the feature specification
P_13	Block inventory: the windows each tier can use
P_14, P_15	Correlation structure per tier, levels and first differences
P_16	Relationship with the target by tier and domain, screened and unscreened
P_17, P_18	Operator scans, full band and diurnal band
P_19	Signed-delay control on the radiation channels
P_20	What radiation adds once temperature is present
P_21	Radiation agreement over the daylight hours
P_22	Radiation and temperature availability, including during target gaps
P_23	The acquisition-fault register, with the evidence per hour
P_F09	Solar radiation as measured, per source, over the on-structure record, with a fortnight enlarged
P_F01	Diurnal radiation by source and season, after alignment
P_F02	Scatter and Bland–Altman, air temperature
P_F03	Mean diurnal cycle by source
P_F05	Correlation and slope against the target, pooled span
P_F06	Correlation heatmaps by tier
P_F07	Slope against the target by tier and domain
P_F08	Operator scans, diurnal band, one figure per tier

The canonical channel reference is `docs/proxy-data-dictionary.md`, generated from the mapping tables in `pc_lib` so that code and documentation cannot drift apart.

Reproduction, from the study directory with `neuralprophet_env` active:

```
jupyter --to ipynb proxy_comparison_study.py
jupyter nbconvert --to notebook --execute --inplace
    proxy_comparison_study.ipynb
```

`studies/unified_dataset/build_unified_dataset.py` must have run first; this study reads its output and the two proxy exports, and nothing else.